



Derivatives

The derivative of a function expresses its rate of change with respect to an independent variable. The first derivative of a function is the slope of the tangent line to that curve defined by the function. The following are the derivatives of different functions: *Excel Review Center*

Algebraic Functions

1. $\frac{dc}{dx} = 0$
2. $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$
3. $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$
4. $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$
5. $\frac{d}{dx}u^n = nu^{n-1} \frac{du}{dx}$ *Excel Review Center*
6. $\frac{d}{dx}\sqrt{u} = \frac{du}{2\sqrt{u}}$
7. $\frac{d}{dx}\left(\frac{u}{c}\right) = \frac{1}{c} \frac{du}{dx}$
8. $\frac{d}{dx}\frac{c}{u} = \frac{-c \frac{du}{dx}}{u^2}$

Exponential Functions

9. $\frac{d}{dx}(a^u) = a^u \ln a \frac{du}{dx}$
10. $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$ *Excel Review Center*

Logarithmic Functions

11. $\frac{d}{dx}(\log_a u) = \frac{\log_a e \frac{du}{dx}}{u}$
12. $\frac{d}{dx}(\log_{10} u) = \frac{\log_{10} e \frac{du}{dx}}{u}$
13. $\frac{d}{dx}(\ln u) = \frac{du}{u}$ *Excel Review Center*

Trigonometric Functions

14. $\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$ *Excel Review Center*
15. $\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$
16. $\frac{d}{dx}(\tan u) = \sec^2 u \frac{du}{dx}$
17. $\frac{d}{dx}(\cot u) = -\csc^2 u \frac{du}{dx}$
18. $\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}$
19. $\frac{d}{dx}(\csc u) = -\csc u \cot u \frac{du}{dx}$

Inverse Trigonometric Functions

20. $\frac{d}{dx}(\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$
21. $\frac{d}{dx}(\cos^{-1} u) = \frac{-1}{\sqrt{1-u^2}} \frac{du}{dx}$
22. $\frac{d}{dx}(\tan^{-1} u) = \frac{1}{1+u^2} \frac{du}{dx}$
23. $\frac{d}{dx}(\cot^{-1} u) = \frac{-1}{1+u^2} \frac{du}{dx}$
24. $\frac{d}{dx}(\sec^{-1} u) = \frac{1}{u\sqrt{u^2-1}} \frac{du}{dx}$
25. $\frac{d}{dx}(\csc^{-1} u) = \frac{-1}{u\sqrt{u^2-1}} \frac{du}{dx}$

Hyperbolic Functions

26. $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$ *Excel Review Center*
27. $\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$
28. $\frac{d}{dx}(\tanh u) = \sec^2 u \frac{du}{dx}$
29. $\frac{d}{dx}(\coth u) = -\csc^2 u \frac{du}{dx}$
30. $\frac{d}{dx}(\sec hu) = -\sec hu \tanh u \frac{du}{dx}$
31. $\frac{d}{dx}(\csc hu) = -\csc hu \coth u \frac{du}{dx}$ *Excel Review Center*

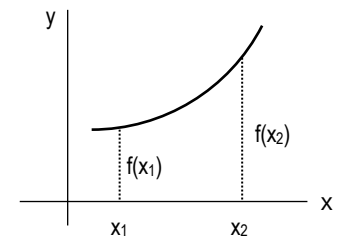
Inverse Hyperbolic Functions

32. $\frac{d}{dx}(\operatorname{arcsinh} u) = \frac{1}{\sqrt{u^2+1}}$
33. $\frac{d}{dx}(\operatorname{arccosh} u) = \frac{1}{\sqrt{u^2-1}}; |x| > 1$
34. $\frac{d}{dx}(\operatorname{arctanh} u) = \frac{1}{1-u^2}; |x| < 1$
35. $\frac{d}{dx}(\operatorname{arc coth} u) = \frac{1}{1-u^2}; |x| > 1$
36. $\frac{d}{dx}(\operatorname{arc sech} u) = \frac{1}{x\sqrt{1-x^2}}; 0 < u < 1$
37. $\frac{d}{dx}(\operatorname{arc csch} u) = \frac{1}{|x|\sqrt{1+x^2}}; x \neq 0$ *Excel Review Center*

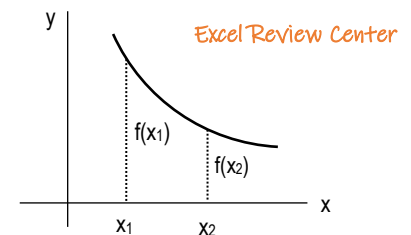
Maxima/Minima Values

If a function f is defined on an interval I , then:

1. f is increasing on I if $f(x_1) < f(x_2)$ whenever x_1, x_2 are in I and $x_1 < x_2$.



2. f is decreasing on I if $f(x_1) > f(x_2)$ whenever x_1, x_2 in I .



3. f is constant if $f(x_1) = f(x_2)$ for every x_1, x_2 in I .

Suppose f is defined on an open interval I and c is a number in I , then:

1. $f(c)$ is a local maximum value if $f(x) \leq f(c)$ for all x in I .
2. $f(c)$ is a local minimum value if $f(x) \geq f(c)$ for all x in I .

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